Integral Calculas:

Single variable $y=f(x)$ the graph is curve

Multi " $z=f(x,y)$ " " " surface

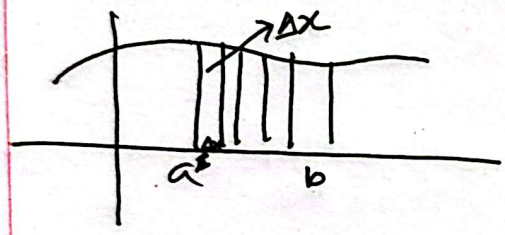
for a circle the $\boxed{\text{method of exhaustion}}$ is equivalent to an equation of the form:

Archimedes

$$\lim_{n\to\infty} A(n) = \pi$$

- here let $A(n)$ denote the area of a regular n-sided polygon inscribed in a circle of radius 1.

rectangle method:



- $\Delta x = \dfrac{b-a}{n}$
- area of n rectangles $\rightarrow A_n = \sum_{i=1}^{n} f(x_i)\Delta x_i$

here the function:

$y = f(x) \geq 0$

it's continuous and non negative and intervals $[a,b]$

$$\boxed{A_n = \int_a^b f(x)\,dx}$$

Reimann's integration

$[n\to\infty$ or $\Delta x \to 0]$

The antiderivative method for areas:

if $A(x)$ Area is under $f(x)$ over $[a,x]$ interval

Then, $A'(x) = f(x)$ $[\because f(x) \geq 0]$

If A dentotes the exact area under the curve and An denotes the approximation to A using n rectangles then

$$\boxed{A = \lim_{n\to+\infty} A_n}$$

$\rightarrow$ rectangle method for computing A

→ linearc integral

$$\int [\alpha f(x) + \beta g(x)]\, dx$$

$$= \alpha \int f(x)\, dx + \beta \int g(x)\, dx$$

Reimann sum, $S_n = \sum_{i=1}^{n} f(x_i^*) \Delta x$

↳ যেকোনো point এর মান

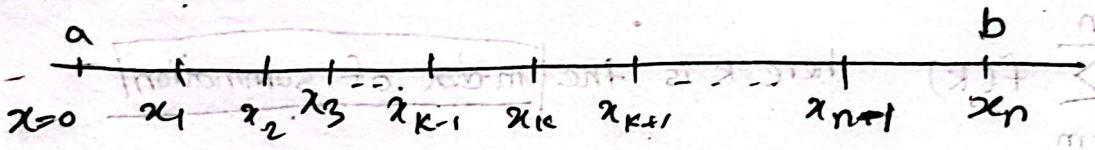
if $f(k)$ is a function of k and if m, n are integers such that $m \leq n$ then,

$$\sum_{k=m}^{n} f(k)$$ here k is the $\boxed{\text{index of summation}}$

[area under a curve]

If the function f is continuous on $[a,b]$ and if $f(x) \geq 0$ for all x in $[a,b]$, then the area A under the curve $y=f(x)$ over the interval $[a,b]$ is defined by,

$$A = \lim_{n \to \infty} \sum_{k=1}^{n} f(x_k^*)\Delta x$$



- left endpoint $x_{k-1} = x_k^* = a + (k-1)\Delta x$ —— (i)
- right " $x_k = x_k^* = a + k\Delta x$ —— (ii)
- midpoint, $x_k^* = a + (k - \frac{1}{2})\Delta x$ —— (iii)

if 'use definition' is mentioned in ques → summation and limit method we have to use.

Theorem

a) $\sum_{k=1}^{n} k = 1+2+\cdots+n = \frac{n(n+1)}{2}$

b) $\sum_{k=1}^{n} k^2 = 1^2+2^2+\cdots+n^2 = \frac{n(n+1)\cancel{(n+2)}(2n+1)}{6}$

c) $\sum_{k=1}^{n} k^3 = 1^3+2^3+\cdots+n^3 = \left[\frac{n(n+1)}{2}\right]^2$